When does data quality help human deferral? Governance-constrained review allocation in knowledge-based decision systems

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Short communication.

Abstract

Automated decision systems must decide which records a model may settle, where scarce review reduces loss, and whether staff can absorb review that governance rules make compulsory. We study a knowledge-based router combining a calibrated predictor, a two-rule data-quality knowledge base, a loss estimator trained on incomplete review logs, and an allocator that enforces capacity and holds mandatory cases it cannot serve. Sorting solves this allocation exactly in feasible and overloaded regimes, and in a multi-period system the backlog does not depend on how mandatory cases are prioritized. We compared learned routers (published learning-to-defer surrogates, differential triage and a two-head estimator) with cost-aware heuristics under matched conditions on four credit datasets, nine data-quality conditions, five simulated reviewer models and 20 splits with corrected tests. Pooled, no learned router beat Chow's cost-risk rule; a cost-sensitive one-vs-all surrogate and the two-head estimator were the strongest learned routers and did not differ significantly. Both beat Chow's rule on all 20 splits on the dataset with about 6,000 training records ($p < 0.001$) and lost with a few hundred, where the two-head estimator was the more data-efficient. Quality inputs reduced cost under outcome-informative missingness on 20 of 20 splits, via estimates of automated rather than reviewer error, but this exploratory effect was not significant after Holm adjustment. In a 40-day simulation, staffing determined backlog and cohort service levels far more than routing did.

Keywords: Knowledge-based systems; Learning to defer; Data quality; AI governance; Human–AI collaboration; Review capacity

1 Introduction

Organizations increasingly let models settle routine credit, claims and eligibility decisions while humans review the rest. Governance frameworks tie this division of labor to accountability and data controls [1, 2, 3, 4], but requiring oversight does not make it effective [5]. The practical question is operational: which records may be automated, where does review reduce expected loss, and can staff absorb the review that data-quality rules make compulsory? The stakes rise when input quality deteriorates, because quality-triggered rules then add mandatory work exactly when capacity is fixed, and cases that cannot be reviewed are neither decided nor resolved.

Existing methods address parts of this problem. Rejection and learning to defer (L2D) trade model error against abstention or expert error [6, 7, 8, 9]; confidence-based wrappers and uncertainty quantification rank cases without measuring review benefit [10, 11]; triage estimates human and model error separately [12]; and recent work adds workload limits [13], sparse expert data [14], quality-aware cascades [15] and congestion [16]. Data-management research measures completeness and validity [17, 18, 19], human-in-the-loop research studies how human effort is deployed [20, 21], and knowledge-based credit decision support does not model reviewer capacity [22]. None of these studies, to our knowledge, combines quality-triggered mandatory review, a capacity that mandatory demand can exceed, and explicit accounting for unresolved cases; nor do they ask whether a quality signal predicts model error, reviewer error, both or neither.

An earlier version of this study reported a 0.2% advantage of quality-aware gain over a feature-aware comparator, using a reviewer simulated from the router's own quality signals and no published baselines. We therefore ask: *under what data-quality and review-capacity conditions does quality-aware deferral reduce decision cost while satisfying explicit governance rules?* The study differs from L2D work in two structural ways: rules make some records non-automatable and some unresolvable, which changes the optimal allocation; and the allocator is separated from the loss estimator, so published surrogates and our estimator compete inside the same governance layer. Contributions: (i) an exact solution of governance-constrained allocation in both regimes, correcting the earlier overload rule, and a proof that backlog is invariant to prioritization; (ii) a matched comparison of learned routers against heuristics under reviewers that do not favor any representation, defects invisible to quality checks and an external dataset; and (iii) a conditions analysis with censoring-aware service measures for managers. We claim neither a new surrogate loss nor a uniformly better router.

2 Method

2.1 Governance-constrained allocation

For record i in a batch of n, a calibrated predictor gives p_i; with costs $c_{FN} = 5$, $c_{FP} = 1$, the automated decision is $\hat{y}_i = \mathbf{1}[p_i \geq 1/6]$ with loss $\ell_i^A = w(y_i)\mathbf{1}[\hat{y}_i \neq y_i]$, $w(1) = 5$, $w(0) = 1$. A reviewer errs with probability e_i, giving expected loss $\ell_i^H = w(y_i)e_i$; review costs $c_R = 0.05$ and an unresolved (held) record $c_H = 1.5$. Observable quality $q_i = (u_i, v_i)$ is the missing-field and out-of-range fraction, with bounds $[Q_{.01,j} - 0.1t_j,\, Q_{.99,j} + 0.1t_j]$, $t_j = \max(Q_{.99,j} - Q_{.01,j}, 1)$, from predictor-training rows. The knowledge base marks record i mandatory, $m_i = \mathbf{1}[u_i \geq 0.5 \vee v_i \geq 0.25]$; thresholds are illustrative organizational policy. With actions $a_i, r_i, h_i \in \{0, 1\}$ (automate, review, hold) and capacity $B = \lfloor bn \rfloor$,

the router solves

$$\min \tfrac{1}{n} \textstyle\sum_i \left[a_i \ell_i^A + r_i(\ell_i^H + c_R) + h_i c_H \right] \tag{1}$$

subject to $a_i + r_i + h_i = 1$, $\sum_i r_i \le B$, $a_i \le 1 - m_i$, $h_i \le m_i$ and the service-first constraint $\sum_i h_i = (|M| - B)^+$, $M = \{i : m_i = 1\}$. Let $g_i = \ell_i^A - \ell_i^H - c_R$.

Proposition 1. *If $|M| \le B$, reviewing all of M plus the $B - |M|$ optional records with the largest positive g_i is optimal. If $|M| > B$, every feasible route reviews exactly B mandatory records and no optional record; reviewing the B smallest ℓ_i^H is optimal. Routes do not depend on c_H.* Proof: in the first case no record is held, so each optional review changes (1) by $-g_i/n$ and an exchange argument applies. In the second, B unheld mandatory records must be reviewed, exhausting capacity; only $\sum_{i \in R} \ell_i^H$ depends on the reviewed set R, and c_H enters as the constant $(|M| - B)c_H$. □ The earlier version ordered overloaded records by estimated gain, which wrongly includes ℓ_i^A for records that cannot be automated.

Proposition 2. *When held records carry over between periods and Proposition 1 is applied each period, holds, backlog, reviews and completions in every period are identical for any ordering of mandatory records.* Proof: by induction, equal backlog implies equal $|M_t|$, hence equal holds $(|M_t| - B)^+$. □ Staffing therefore sets the backlog, approximately growing by $(\mu_t - b)^+ n$ per period for mandatory rate μ_t, whereas prioritization only decides who waits.

2.2 Loss estimation

The allocator accepts any estimates of ℓ_i^A and ℓ_i^H. Our two-head estimator fits two gradient-boosted regressors on the eight inputs $z_i = (p_i, s_i, q_i, p_i q_i, \hat{y}_i q_i)$, where $s_i = \min(5p_i, 1 - p_i)$ is the Bayes risk of automation: an *automation head* $\hat{\ell}^A(z)$ fitted on all labeled records, and a *reviewer head* $\hat{\ell}^H(z)$ fitted only on records with one logged review decision, using the unbiased target $w(y_i)\mathbf{1}[\text{review} \ne y_i]$. Optional records are reviewed only if $\hat{g}_i = \hat{\ell}^A - \hat{\ell}^H - c_R > 0$; overloaded mandatory records are ordered by $\hat{\ell}^H$. Allocation costs $O(n \log n)$; each decision is logged as a label-free audit tuple $(p_i, q_i, m_i, \hat{\ell}_i^A, \hat{\ell}_i^H, \text{action})$, which documents routing rather than explaining the model [23, 24, 25].

3 Experimental protocol

We use the credit benchmarks [26] German (1,000 records), Australian (690) and default (30,000) [27, 28, 29], and HMEQ (5,960; natural missingness in 11 of 12 predictors) as an external dataset unused during development [30]. Feature-identical records are grouped and split by majority label into 45% predictor training, 15% calibration, 20% gain fitting and 20% testing over seeds 2–21. A histogram gradient-boosting classifier (120 iterations, 15 leaves)

is calibrated by logistic regression; loss heads are boosted regressors (100 iterations, seven leaves); calibration and gain partitions are expanded into clean, missing, outlier and mixed copies.

Test sets face nine conditions: clean; Family A, defects visible to quality checks (missing, outliers, mixed, and a held-out severe shift with 35–75% missing cells); and Family B, domain-valid defects never used in fitting (outcome-informative missingness, stale records with continuous fields deflated by 10–45%, inconsistent but valid category codes, and a realistic mix). Five reviewer models are simulated: constant ($e = 0.10$), quality-coupled ($\min(0.4, 0.05 + 0.35u + 0.25v)$, the earlier design), feature-dependent (a hidden projection of clean features), AI-correlated (errors concentrated where the model errs) and adverse (0.25); only the second uses the router's signals. Learned methods train on noisy, incomplete logs: 40% of the expanded gain-fitting rows carry one sampled reviewer decision.

All methods share the predictor, rules, costs, capacity and allocator. Comparators are confidence ($-|p_i - 1/6|$), Chow cost risk (s_i) [6], quality only ($u_i + v_i$), the earlier single-head gain, L2D-CE with cost-sensitive weights including the review fee [8], L2D-OvA [9] trained as published and routed by a cost-sensitive plug-in rule on its calibrated outputs (expected automation loss minus expected review loss minus c_R), differential triage [12] and a probability-only two-head ablation; L2D models are one-hidden-layer rejectors. Heuristics fill capacity; learned methods review only positive scores. As splits reuse observations, tests use the corrected resampled t-test [31] with Holm adjustment [32] within declared families: pooled comparisons (8 tests), per-dataset comparisons (9), data-volume comparisons (7) and Table 2 (18, exploratory). A 40-day simulation (100 arrivals per day, severe shift on days 11–20, held cases carried forward at 0.25 per case-day, c_H for cases open at day 40) measures service per arrival cohort, counting cases unresolved at day 40 as unmet and estimating resolution-time quantiles by Kaplan–Meier with censoring.

4 Results and discussion

Heuristics lead in pooled data; learned routers need data. All rule-constrained methods reviewed 19.8% and held 6.0% of records with zero violations, so cost differences reflect which records were reviewed (Table 1). Pooled, Chow (0.5291) and confidence (0.5295) were cheapest; the best learned routers, L2D-OvA (0.5312) and the two-head estimator (0.5355), did not differ significantly (14/20 splits favored OvA; $p = 0.48$). The two-head router beat L2D-CE by 1.95% (19/20; Holm $p = 0.11$) and trailed Chow by 1.2% (5/20; Holm $p = 0.70$). Chow or confidence was cheapest under four of five reviewer models, OvA under the feature-dependent one. Dataset size explains the pattern: with about 6,000 gain-fitting records (default data), OvA and the two-head router beat Chow by 1.4% and 1.1% on all 20 splits (Holm $p < 0.001$); with 140–200 records (German, Australian) both lost. Subsampling the

default partition (Fig. 1) left the two-head router 2.0% costlier than Chow at about 200 records, approximately competitive at 1,800 (+0.1%, 10/20, $p = 0.85$) and cheaper only at 6,000 (−1.1%, 20/20). It was the most data-efficient learned router, beating OvA on all 20 splits at 200 and 600 records (by 3.0% and 2.7%; Holm $p < 0.005$), whereas OvA was cheapest at 6,000 (−1.4%). This shows data dependence in one dataset, not a deployment threshold.

Table 1: Expected cost per record at 20% capacity (lower is better), averaged over eight non-clean conditions, five reviewer models and 20 splits; HMEQ is evaluated on its natural data. Learned methods are trained on noisy, incomplete review logs. Bold: lowest deployable cost per column.

Method	German	Australian	Default	Macro	HMEQ (natural)
Automation only (rules ignored)[a]	0.6254	0.4046	0.6291	0.5530	0.2583
Random	0.6327	0.4752	0.6012	0.5697	0.2700
Confidence	0.6045	**0.4139**	0.5700	0.5295	**0.1847**
Chow cost risk [6]	**0.5987**	0.4204	0.5684	**0.5291**	0.2288
Quality only	0.6514	0.4807	0.5901	0.5741	0.2960
Single-head gain (earlier version)	0.6123	0.4329	0.5634	0.5362	0.2065
L2D-CE [8]	0.6219	0.4469	0.5695	0.5461	0.2182
L2D-OvA [9]	0.6086	0.4247	**0.5604**	0.5312	0.2103
Differential triage [12]	0.6108	0.4352	0.5657	0.5372	0.2082
Two-head, probability only	0.6100	0.4338	0.5705	0.5381	0.2163
Two-head, quality-aware (proposed)	0.6104	0.4338	0.5622	0.5355	0.2053
Oracle (uses labels)[b]	0.4724	0.2904	0.3692	0.3773	0.0624

[a]Ignores the rules (12.6% of records automated in violation). [b]Uses true labels and reviewer losses; an optimistic lower bound on attainable cost, not deployable.

Quality signals act on automated error (exploratory). Adding quality inputs to one head at a time (Table 2) shows that under outcome-informative missingness they reduced cost through the automation head (2.8%, 20/20 splits) but not the reviewer head; HMEQ's natural missingness showed the same pattern (+4.5%, 17/20, versus +1.0%). This is consistent with incompleteness predicting the model's own error, but no cell survives Holm adjustment, so the mechanism awaits confirmatory testing. Through the reviewer head, quality inputs helped only when reviewer error was built from quality (+0.7%, 16/20), so the earlier simulation favored the representation slightly. Effects for stale records and miscoded categories, invisible to quality checks, were small. Ordering overloaded records by $\hat{\ell}^H$ rather than gain lowered cost by only 0.18% ($p = 0.47$), and two heads beat a single head only directionally (+0.6% and +0.5% with 10% and 40% of rows logged, −0.5% with complete logs; all $p > 0.5$).

Rules protect only against visible defects. Under severe shift the rules forced 47.5% of records into holds and raised cost by 67% relative to an unconstrained router, which automated 54% of policy-barred records. Stricter

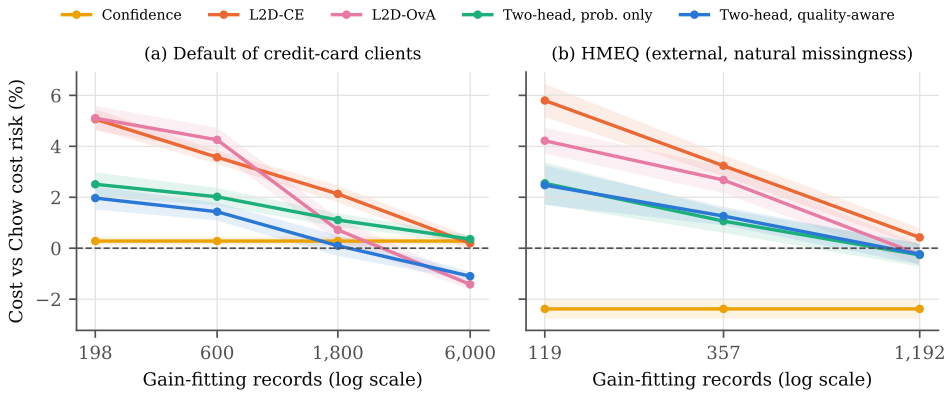


Figure 1: Learned routing versus data volume on (a) default and (b) HMEQ data. Cost relative to Chow's rule as the gain-fitting partition grows (20% capacity, 20 splits per size); negative values favor the learned router. Bands: ± 1.96 standard errors across splits.

Table 2: Value of quality inputs by estimator head: cost reduction (%) relative to the probability-only two-head router at 20% capacity, with splits won out of 20 in parentheses.

Condition	Automation head only	Reviewer head only	Both heads (proposed)
Informative missingness (B)	+2.83 (20)†	−0.28 (7)	+2.29 (18)†
Stale records (B)	+0.98 (13)	+0.04 (10)	+0.91 (14)
Inconsistent codes (B)	+0.15 (10)	−0.23 (7)	−0.08 (10)
Visible defects (A), quality-coupled reviewer	+0.07 (13)	+0.70 (16)	+0.65 (16)
Visible defects (A), four decoupled reviewers	+0.18 (11)	−0.24 (5)	−0.07 (10)
HMEQ, natural missingness (external)	+4.53 (17)†	+0.95 (14)	+5.12 (19)†

Bold: Holm-adjusted $p < 0.05$ over the 18 cells (none). †Unadjusted corrected $p < 0.05$ only. Exploratory analysis.

thresholds cut the share of truly defective records automated from 80% to 4.5% under visible shift but left 75% automated under invisible defects, so zero violations do not establish adequate control. Re-optimizing every method for cost ratios of 2, 5 and 10, review fees of 0–0.2 and hold penalties of 0.5–3.0 kept confidence cheapest in all seven settings, with the best learned router within 0.7% of Chow; hold penalties shifted cost linearly with identical routes, as Proposition 1 predicts.

Staffing, not routing, sets service levels. Mandatory demand rose from 5.5% to 70% of arrivals during the simulated episode (Fig. 2). As Proposition 2 predicts, all prioritization rules produced identical backlogs, and the fluid estimate matched peaks of 500, 300 and 200 records at 20%, 40% and 50% capacity. At 20% capacity 21.7% of mandatory cases remained unresolved at day 40 and only 36% of each arrival cohort was resolved within two days, although 48% of eventually resolved cases met that target; the 95th-percentile resolution time was never

reached. At 40% capacity all cases were resolved and cost per arrival fell by 45%. Prioritization only redistributed waiting: at 40% capacity first-in-first-out cut the Kaplan–Meier 95th-percentile resolution time from 13.9 to 7.1 days and raised seven-day resolution from 76% to 94%, but lowered two-day resolution from 65% to 40%. Managers should therefore validate any learned router against a cost-aware heuristic on local data, staff to mandatory rather than average demand, choose the queue discipline for its service profile, and report cohort service levels and defect coverage, not violations alone [33, 34].

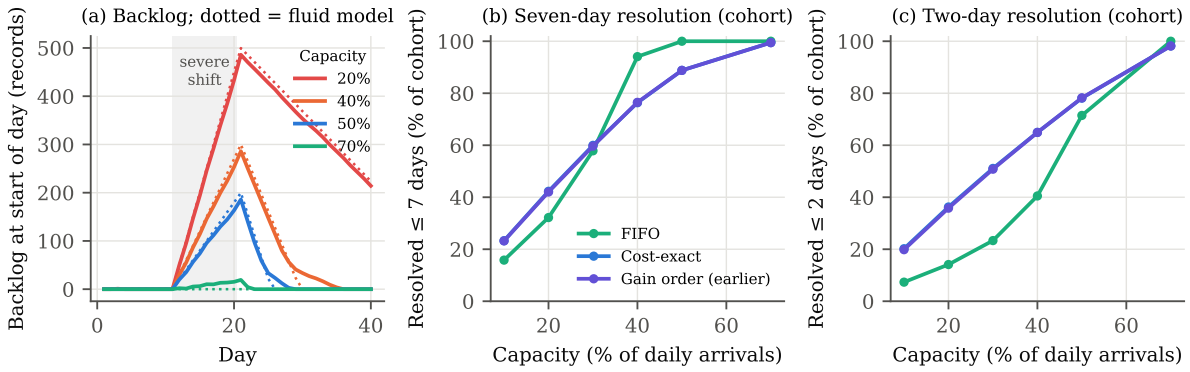


Figure 2: Backlog and cohort service under a severe-shift episode (days 11–20). (a) Backlog at four staffing levels; dotted lines are the fluid approximation. (b) Seven-day and (c) two-day resolution per arrival cohort, counting cases unresolved at day 40 as unmet, by prioritization rule (four datasets, two reviewers, ten seeds).

5 Limitations

Reviewers are simulated; real reviewers may anchor on model output or tire under backlog, so validation with review logs is the main next step. Benchmarks are old, Family B defects stylized, gradual drift unmodeled [35], and thresholds and costs illustrative. The quality mechanism and three of four prespecified hypotheses are inconclusive, L2D baselines are adapted rather than tuned, data-volume evidence comes from one dataset, and fairness and legal sufficiency are not assessed.

6 Conclusion

Separating rules, loss estimation and capacity makes governance-constrained routing exact and auditable for any estimator. No learned router beat a cost-aware heuristic in pooled data; learned routers helped where review history was large, and quality signals appeared useful under informative missingness pending confirmation. Under deterioration, staffing, not routing, determined service completion.

Data and code availability Code, seed-level results, tables, figures and tests: `https://github.com/ykunam/data-quality-budgeted-deferral` (directory `budgeted_deferral`). Datasets: [27, 28, 29, 30]. A versioned archival release (DOI) must be added before submission.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process During the preparation of this work the author used ChatGPT (OpenAI) for study design, code and drafting of the original version, and Claude (Anthropic) to reimplement and extend the experiments, implement baselines, analyze results, generate the two data figures with analysis code, check literature and draft this revision. [After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.]

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